DIVYANSHU GOYAL

Multimodal Foundation Models | RL Post-Training | Agentic Evaluation

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SUMMARY

AI researcher and engineering leader with **8+ years of experience** specializing in **agentic AI and Multimodal LLMs**. I bridge the gap between cutting-edge research and enterprise-scale products. During my tenure at Adobe, I led the technical strategy to scale **GenStudio for Performance Marketing from 0 to 1**, driving **multi-million-dollar adoption** across the Fortune 500. Currently, I am **tech lead for a 7-person team** building specialized VLMs, judge models, and retouchers. Combining product execution with rigorous research, my work includes co-leading the ATune photo retoucher (**outperforming GPT-6 Astra** in human preference), **6 patents**, and publications at CVPR, ICML, and AAAI workshops.

EXPERIENCE

Applied Scientist – Adobe Inc., San Jose, CA June 2021 – Present

FixIt: Multimodal Models for Marketing & Visual Quality (Tech Lead, Team of 7)

- Architected a **sub-second** JEV-style judge model integrated directly into the **Firefly AI Agent evaluation harness**, improving the agent's outputs through **real-time feedback** and reducing user revisions by **40%**.
- Directed the development of **three specialized model streams**: domain-adapted VLMs, marketing-quality judges, and an **unsupervised photo retoucher** trained entirely without human-edited pairs.

ATune: Unsupervised Photo Retouching (Co-Lead with Adobe Research)

- Co-led project architecture and evaluation design with Adobe Research as co-first author, owning the synthetic data pipeline, training recipe, and human evaluation framework, and **mentored a PhD intern** on the project.
- Trained a compact **54M-parameter model** that performs non-destructive photo adjustments via off-the-shelf editors (RawTherapee/darktable), learned from **10M unedited photographs with zero expert targets**.
- In a 157-rater human study (3,524 judgments), **beat GPT-6 Astra** (54.3% of decisive votes, 95% CI 50.3–58.3) and **matched Claude Opus 5** (50.5%); preferred over GPT-6 Astra by $\geq$8 of 9 published quality judges across FiveK, PPR10K, and SPAQ.
- Ran data-scaling studies from 1M to 10M images: each larger corpus wins or ties, and no judge on any test set prefers a smaller one.

Adobe Marketing VLM (Gemma-3-12B, SFT & GRPO Alignment)

- Fine-tuned **Gemma-3-12B** via distributed SFT followed by multi-round GRPO on $8\times$ H100s, leveraging LoRA at rank 256 (1.19B trainable parameters) across 7 custom reward functions.
- Identified **reward exploitation** in ordinal rating prediction: the policy collapsed toward the middle rating (84% of predictions vs. 56% in human annotations), exploiting the bounded distance-based reward. Introduced a **cubic distance penalty** and **class-weighted advantages** for underrepresented ratings, reducing middle-rating predictions to **64%**.
- Built a **fault-tolerant** annotation pipeline that labeled $\sim$**10M images** with a teacher VLM to create the SFT dataset.
- Built a **Photoshop agent** powered by the marketing VLM that edits photos toward **marketing taste**, turning the model's judgment of what makes an image effective into concrete Photoshop edits.

JEV-Style Marketing Judge Models

- Trained **Qwen3.5-4B** LoRA judges that output **calibrated scores in a single forward pass** instead of generating text, on a **1.1M soft-labeled dataset** curated from 14 human-rated sources, raising KonIQ correlation to **0.663 SRCC** (vs. 0.421 for the reference model).
- Curated a **67K-item held-out benchmark**, leak-checked against all training images via SSCD copy detection, and **gated every model release** on a pre-registered non-inferiority test, so a new model shipped only if it held its quality on unseen data.
- Conceived and built a **live demo suite** that turns the judge from a scorer into a **creative engine**: it picks the best crop of an image (**0.82 pairwise accuracy on GAIC** vs. 0.71 for the score-based baseline), guides photo retouching, scores marketing and custom attributes, re-ranks Adobe Stock for marketing use, and flags low-quality assets. Presented it to **leadership and external partners**, opening new customer engagements.
- Diagnosed a **vision-encoder collapse** that left every checkpoint image-blind within 200 steps while the loss looked normal, and added an automated **vision-health check** that halts training on collapse, catching silent failures the loss misses.

Adobe GenStudio for Performance Marketing (Tech Lead, Team of 20+)

- Took GenStudio from **prototype to a multi-million-dollar enterprise product** (launched Oct 2024), leading its technical strategy and the production API layer that serves its LLM-powered features to **Fortune 500** customers.
- Co-invented automated brand-guideline extraction and compliance scoring (US20250117829), the foundational engine for **Adobe Brand Intelligence**.
- Developed parameter-efficient fine-tuning pipelines on a Fortune 500 customer's brand data, directly closing a **custom-model enterprise deal**.

Evaluation Research

- **DistortBench** (First Author; CVPR/ICML 2026 Workshops): Established a 13.5K-question visual distortion benchmark across 27 types and 5 severity levels; the best of 18 VLMs (61.9%) **trails a human majority vote** (65.7%).

Machine Learning Engineer – Adobe Inc., Bangalore, India July 2016 – August 2019

- Joined Adobe's core ML serving platform team at inception and helped plan the end-to-end serving infrastructure for internal workloads, including field-aware factorization machines.
- Scaled production inference throughput to **3,500 QPS per node at 0.3–0.6 ms p99 latency** (Adobe Tech Blog), implementing a garbage-collection-free asynchronous event pipeline on LMAX Disruptor.
- Built the model lifecycle platform for versioned deployment, monitoring, and staged rollout.

PUBLICATIONS

ATune: Unsupervised Photo Retouching Through Image-Editor Controls
D. Goyal*, G. Ruiz*, A. Meisami | *Under review* | *Equal contribution

DistortBench: Benchmarking Vision Language Models on Image Distortion Identification
D. Goyal, A. Eppa, V. Bannihatti Kumar | *CVPR & ICML 2026 Workshops* | arXiv:2604.19966

JudgeBreak: Is Taste Hidden in the Layers? Stress-Testing Reward Models and LLM Judges with Representation-Based Subjective Evaluation
V. Bannihatti Kumar, A. Narang, M. Anand, A. Eppa, **D. Goyal** | *ICML 2026 Workshop* | OpenReview

Curiosity-Driven LLM-as-a-Judge for Personalized Creative Judgment
V. Bannihatti Kumar, **D. Goyal**, A. Eppa, N. Bhandari | *AAAI 2026 Workshop (**Oral**)* | arXiv:2510.05135

PATENTS

Title	Number	Year
Guiding Language Translation with Translation Documents Using ML	US20260050753	2026
Brand-Aligned Marketing Content Generation (Adobe Brand Intelligence)	US20250117829	2025
Cultural Adaptation of Images Using Generative AI	US20260051084	2026
Parameter-Efficient Fine-Tuned Models	US20250217193	2025
Video-to-Video Culturalization Using Generative AI	Filed with USPTO	2026
Learning to Be Surprised: A Curiosity-Driven Framework for Creative Judgment	Filed with USPTO	2026

EXTERNAL CONTRIBUTIONS

VibeNanoChat: GPT-2 Scale LLM Pretraining from Scratch 2026 | code
Pretrained GPT-2-scale transformers from scratch at **>1M tokens/s** on 8× H100 using a custom distributed Muon optimizer; derived empirical compute-optimal scaling frontiers from isoFLOP profiles over 40+ runs. *Technical blogs:* Going Fast, Scaling Transformers.

BrownBearAI: On-Device Photo Understanding for iOS 2026 | App Store
Shipped a privacy-first iOS app executing on-device neural face clustering and deduplication at **99.4% BCubed F1** on LFW identities with zero cloud dependency.

LaSeRo: Segmenting Antarctic Rock Outcrops from Landsat 8 2019–2020 | code
Trained and evaluated SegNet architectures to segment geological rock outcrops across multispectral satellite imagery.

Face Mask Detection for COVID-19 2020 | code
Developed a real-time mask detection pipeline, benchmarking CNN architectures with VAE representation learning and Grad-CAM visual attribution.

TECHNICAL SKILLS

Research: VLM Training & Evaluation, Supervised Fine-Tuning (SFT), Reinforcement Learning (GRPO), Reward Design, LLM-as-a-Judge & Reward Models, LLM Pretraining & Scaling Laws, Benchmark Design, Human Preference Studies

Tools: Python, PyTorch, vLLM, Distributed Training (DDP, ZeRO-2), torch.compile, FlashAttention-3, Liger Kernels, Muon, Mixed Precision (BF16), LoRA, Hugging Face Transformers/PEFT, Weights & Biases

EDUCATION

Institution	Degree	GPA	Years
Georgia Institute of Technology	M.S. Computer Science (Machine Learning)	4.0/4.0	2019 – 2021
BITS Pilani	B.E. Computer Science, M.Sc. Mathematics	8.75/10	2011 – 2016